

LECTURE 02

MULTIVARIABLE CALCULUS

EXAMPLE 01

$$y = -x^2 + 4x$$

Since $a = -1 < 0$ parabola is opening downward vertex occurs at $x = -\frac{b}{2a} = \frac{-4}{2(-1)} = 2$

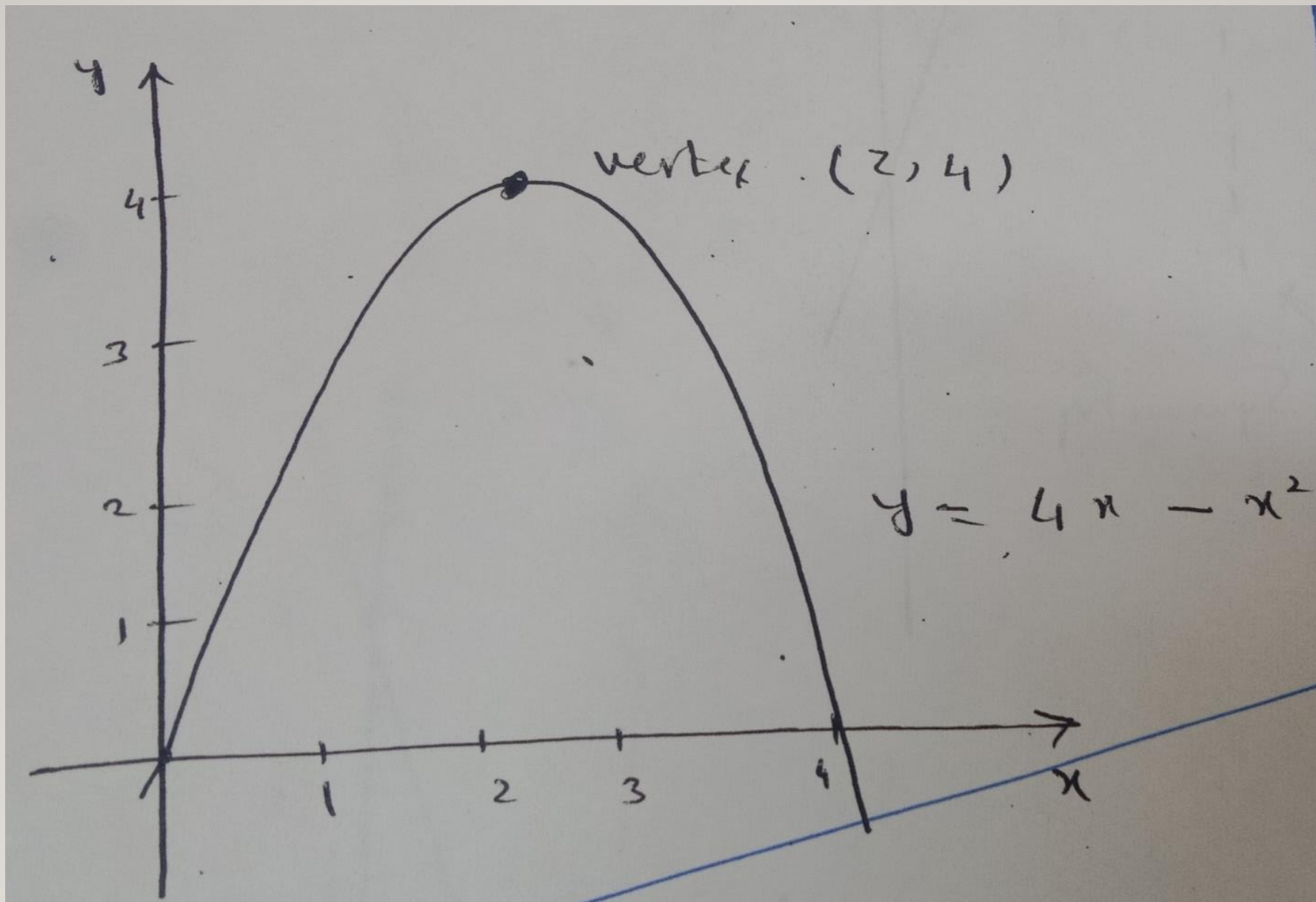
Axis of symmetry is the vertical line $x = 2$. The y-coordinate of the vertex is

$$y = -(2)^2 + 4(2) = 4$$

Hence vertex is $V(2,4)$. The zeros of the parabola (i.e., the point where the parabola meets x-axis) are the solutions to $-x^2 + 4x = 0$

$$\Rightarrow x = 0 \text{ and } x = 4$$

Therefore $(0, 0)$ and $(4, 0)$ lie on the parabola. Also $(1, 3)$ and $(3, 3)$ lie on the parabola.



EXAMPLE

$$y = x^2 - 4x + 3$$

Since $a = 1 > 0$ parabola is opening upward vertex occurs at $x = -\frac{b}{2a} = \frac{4}{2} = 2$

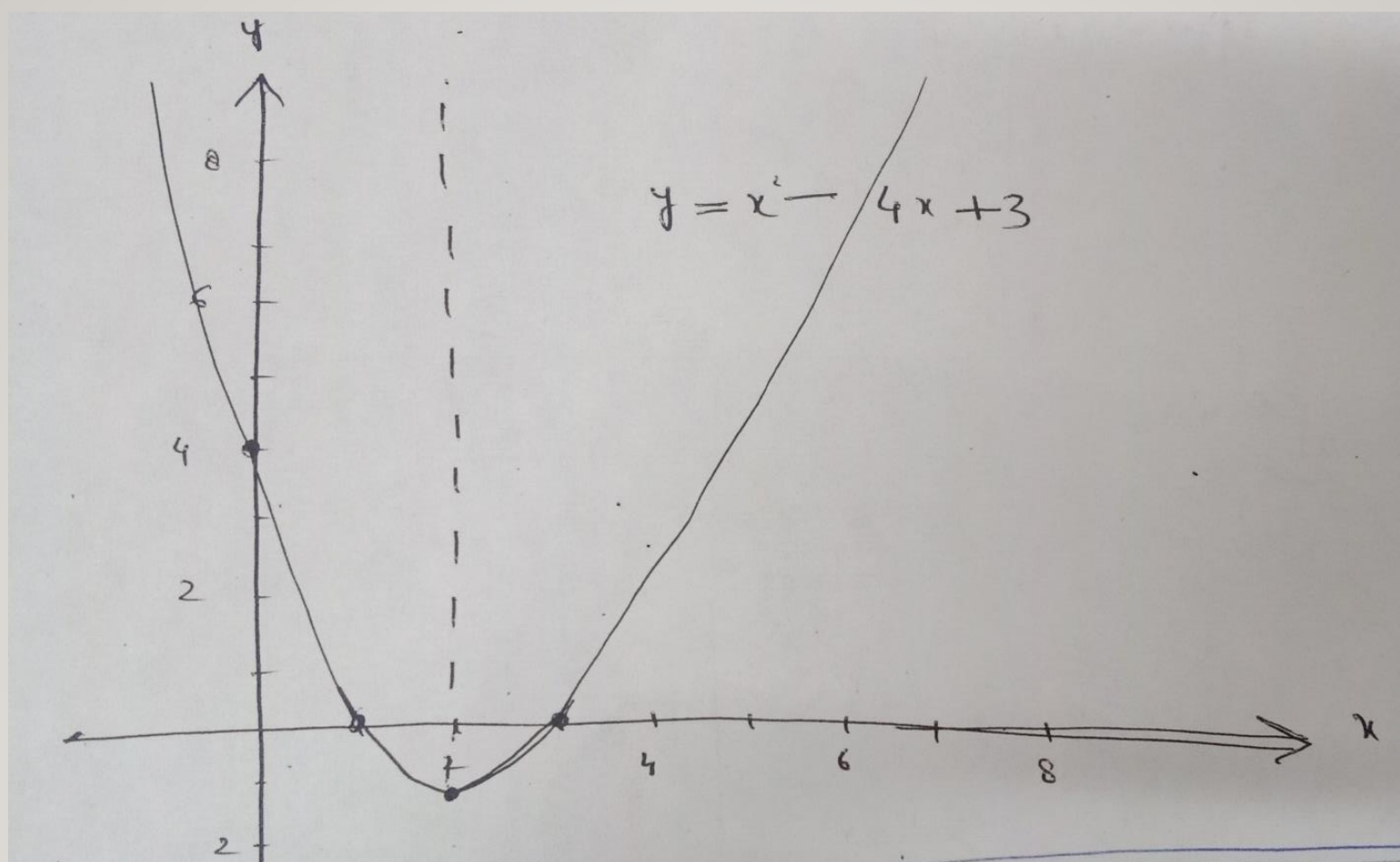
Axis of symmetry is the vertical line $x = 2$. The y-coordinate of the vertex is

$$y = (2)^2 - 4(2) + 3 = -1$$

Hence vertex is $V(2, -1)$. The zeros of the parabola (i.e., the point where the parabola meets x-axis) are the solutions to $x^2 - 4x + 3 = 0$

$$\Rightarrow x = 1 \text{ and } x = 3$$

Therefore $(1, 0)$ and $(3, 0)$ lie on the parabola. Also $(0, 3)$ and $(4, 3)$ lie on the parabola.



EXERCISE:

SKETCH THE PARABOLAS

1) $y = 4x^2 + 8x + 5$

2) $y = 1 - 4x - x^2$

3) $y^2 = 6x$

4) $y^2 = -10x$

5) $x^2 = -9y$

6) $x^2 = 4y$

7) $5y^2 = 12x$

8) $x^2 - 40y = 0$

9) $x^2 - 4x + 2y = 1$

10) $y^2 - 6y - 2x + 1 = 0$

11) $x = y^2 - 4y + 2$

12) $-y^2 + 2y + x = 0$

ELEMENTS OF THREE DIMENSIONAL GEOMETRY

Distance formula in Two Dimension: Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be two points in plane the distance between them is

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Distance formula in Three Dimension: Let $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ be distinct point in space such that PQ is not parallel to one of the coordinate axes then

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

MID POINT OF TWO POINTS

If R is the middle point of the segment PQ the coordinates of the middle point are

$$x = \frac{x_1 + x_2}{2}, \quad y = \frac{y_1 + y_2}{2}, \quad z = \frac{z_1 + z_2}{2}.$$

Example: Let us consider two points $A(3, 2, 4)$ and $B(6, 10, -1)$ then the midpoint C of AB has coordinates

$$\left(\frac{3 + 6}{2}, \frac{2 + 10}{2}, \frac{4 - 1}{2} \right) = \left(\frac{9}{2}, 6, \frac{3}{2} \right)$$

EXERCISE

Find the distance between them and the midpoint of the line segment joining them.

a) $P(0,0,0); Q(2,1,3)$

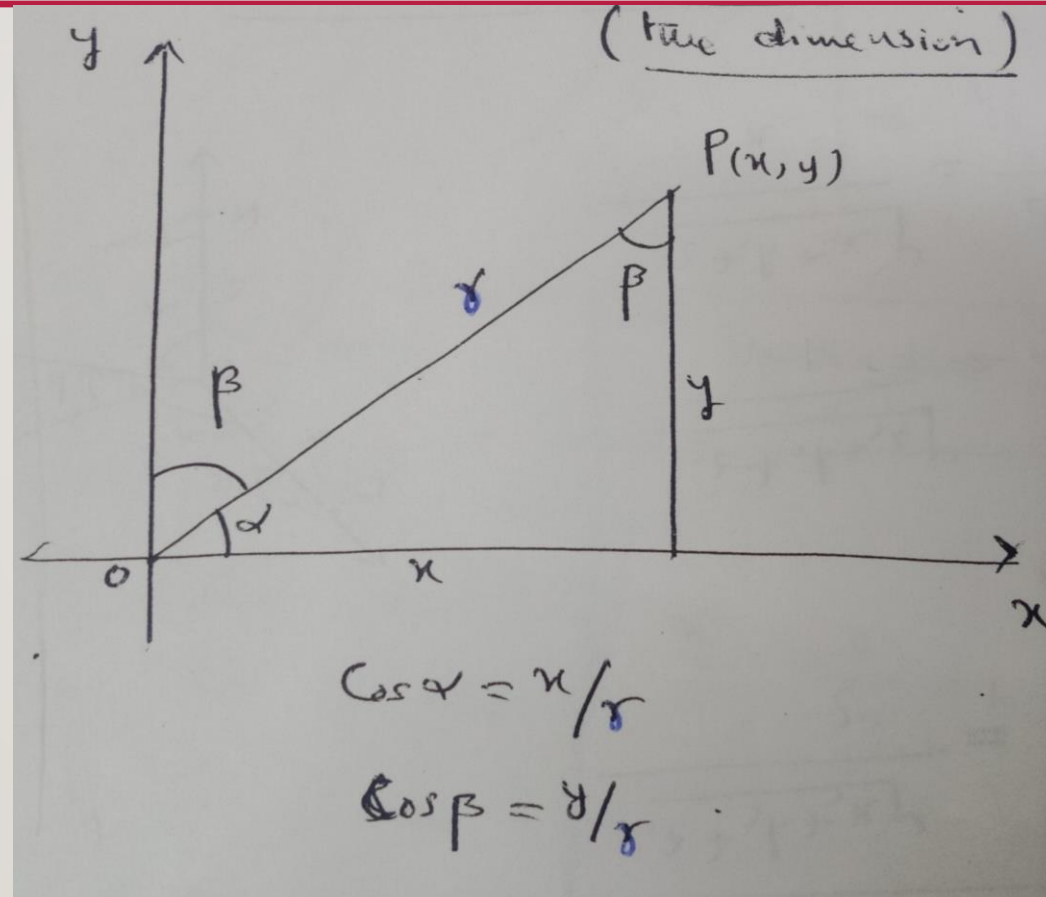
b) $P(5,2,3); Q(4,1,6)$

c) $P(-2, -1, 3); Q(3, 0, 5)$

d) $P(-1, -1, -3); Q(4, 3, -2)$

e) $P(4, -1, 3); Q(2, 3, -1)$

GIVEN A POINT FINDING ITS DIRECTION COSINES



DIRECTION ANGLES (THREE DIMENSION)

The direction angles α, β, γ of a line is defined as

$\alpha = \text{Angle between line and the positive } x - \text{axis}$

$\beta = \text{Angle between line and the positive } y - \text{axis}$

$\gamma = \text{Angle between line and the positive } z - \text{axis}$

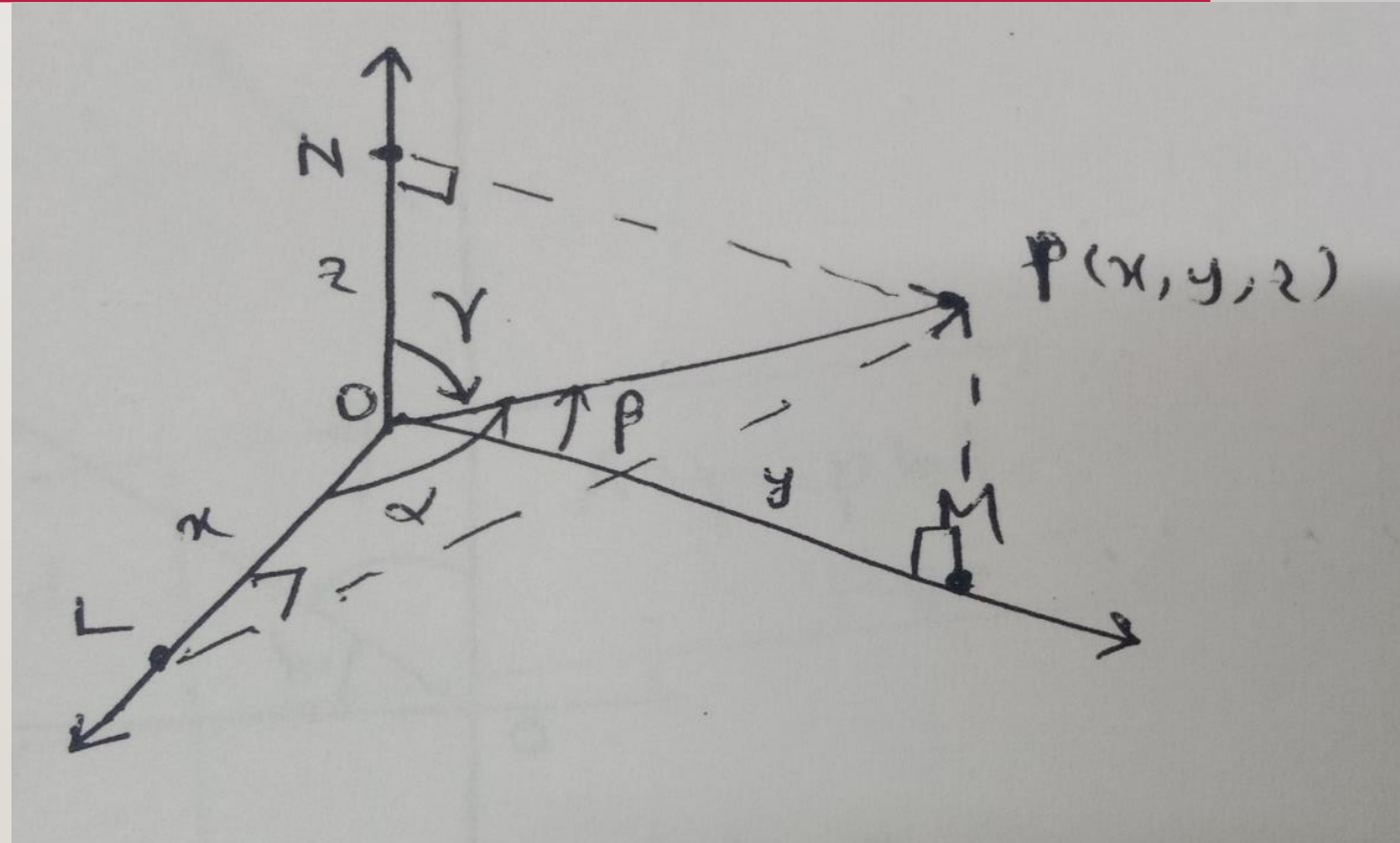
By definition each of these angles lies between 0 and π .

Direction Ratios

- Cosines of direction angles are called direction cosines
- Any multiple of direction cosines are called direction numbers or direction ratios of the line L.

DIRECTION ANGLES OF A LINE

$$\cos \alpha = \frac{x}{OP} = \frac{x}{\sqrt{x^2 + y^2 + z^2}}$$
$$\cos \beta = \frac{y}{OP} = \frac{y}{\sqrt{x^2 + y^2 + z^2}}$$
$$\cos \gamma = \frac{z}{OP} = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$



PROVE THAT $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

Proof:

$$\begin{aligned}\cos^2\alpha + \cos^2\beta + \cos^2\gamma &= \frac{x^2}{OP^2} + \frac{y^2}{OP^2} + \frac{z^2}{OP^2} \\ &= \frac{x^2 + y^2 + z^2}{OP^2}\end{aligned}$$

where, $OP^2 = x^2 + y^2 + z^2$

$$\Rightarrow \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$